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Project Title:	Long Term Space Flight Impacts on Male Reproductive Health		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
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Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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COI Name (Institution):	Roby, Katherine (University of Kansas Medical Center) Gupta, Vijayalaxmi (University of Kansas Medical Center) Holets, Lesya (University of Kansas Medical Center)		
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Task Description:	Published data demonstrates that short term microgravity (μG) (~2 wks) produces small but significant reductions in spermatogenesis in rats. However, long term ground-based μG models using the rodent hindlimb unloading (HLS) (> 3 wks), with surgical prevention of cryptorchidism, show dramatic testicular atrophy and loss of spermatogenesis leading to sterility. These results and related literature in humans suggest that longer duration space flight may have negative impacts on spermatogenesis and compromise male health and increase cancer risk. Thus, the testable hypothesis driving this proposal is: Exposure to μG 3 wks or longer causes disruption of spermatogenesis in mice. This proposal addresses the Bioastronautic Roadmap (BR) Risk 30: Chronic and Degenerative Tissue Risks in that long term μG-induced testicular atrophy will increase the risk for long term health problems post-flight including increased cancer risk and sterility. The hypothesis will be tested with the following specific aims: 1. Identify specific histopathology and dysfunction in spermatogenesis and testis function in mice exposed to Low Earth Orbit (LEO) for 30 d (as determined by mission duration) and compare to ground-based controls. 2. Quantify declines in sperm motility and morphology in cauda epididymal sperm from male mice exposed to LEO for 30 d (as determined by mission duration) and compare to ground-based controls. 3. Determine dysregulation in gene transcription of markers of spermatogenic regulation and inflammatory markers in testes from mice exposed to LEO for 30 d (as determined by mission duration) and compare to ground-based controls. 4. Determine declines in fertility and fecundity of male mice exposed to μG (This aim will depend upon availability of flight animals for mating trials). The opportunity to expose male mice for 30 d of μG in the new BION flights represents the first to directly evaluate this reproductive health problem directly in long term μG under properly controlled conditions. These experiments will validate the ground-based long term HLS rodent model and provide proof that male reproductive health is put at risk by extended exposure to μG. The experiments will provide an initial data set for development of appropriate countermeasures to the negative impact of long term space flight on endocrine and reproductive physiology compromising male health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Female reproductive studies: The observed down-regulation of estrogen receptor ERa, combined with the histological changes, suggests that the animals stopped cycling in space flight and estrogen support was severely impaired. The observations are also significant in that the results were verified on three separate Space Shuttle experiments (STS-131, STS-133, and STS-135). If ERa is also down regulated in other tissues, this would indicate a common mechanism underlying space-flight induced degradation of other estrogen dependent systems including bone loss, muscle weakness, immune response, and wound repair. A broad effect of space flight on ERa receptor levels and/or signaling will offer a common pathway to generate countermeasures for space flight. Since similar changes occur in the same organ systems during the aging process, the countermeasures would have application for improving quality of life in the same organ systems during the aging process. Male reproductive health: The experiments outlined in this proposal will be the first to determine directly, under controlled space flight conditions, whether exposure to long duration microgravity (μG) disrupts sperm production and compromises male reproductive health in mammals. The unique opportunities offered in the new BION 30 d+ rodent flights are the first to extend space flight beyond all previous μG exposures for rodent models on any flight platform. As such, the planned experiments will be the first to directly assess a long term health risk that has been identified by ground-based space flight animal and human models. Thus the hypothesis driving the research: that long term space flight has negative impacts on male reproductive health. Specific Aim 1. Identify specific histopathology and dysfunction in spermatogenesis and testis function in mice exposed to LEO for 30 d (as determined by mission duration) and compare to ground-based controls. Specific Aim 2. Quantify declines in sperm motility and morphology in cauda epididymal sperm from male mice exposed to LEO for 30 d (as determined by mission duration) and compare to ground-based controls. Specific Aim 3. Determine dysregulation in gene transcription of markers of spermatogenic regulation and inflammatory markers in testes from mice exposed to LEO for 30 d (as determined by mission duration) and compare to ground-based controls. Space flight impacts on reproductive health include both microgravity and radiation environmental factors. Space flight changes are an excellent accelerated analogue to the aging process on Earth. The 30 day BION M1 mission will combine the longest exposure and the largest number of mice in a single mission and represent a significant duration relative to the complete spermatogenesis cycle in the testis. Changes in sperm motility, spermatogenesis, gene transcription, and DNA damage in long term space flight will offer the first comparison for identical male reproductive health outcome measures for the normal aging process on Earth.
	The BION M1 flight was the first opportunity to directly test the hypothesis that prolonged space flight negatively affects testicular function and spermatogenesis. Seven mature male mice were flown in orbit for 30 d. Asynchronous diet and age-matched ground control (GC) mice were maintained in the same BION hardware on the ground for 30 d. For histology, testis and epididymis were fixed in Bouin's solution, washed in 70% ETOH, paraffin-embedded, and processed for hematoxylin and eosin staining. For RNA and protein extraction tissues were harvested in RNAlater and stored at -80°C. RNA was subjected to real-time RT PCR. Next Generation sequencing was used to compare gene expression profiles between flight and GC mouse testes. We report preliminary results from analysis of a subset of flight and GC animals for histological modifications, and testicular/spermatogenesis gene expression changes. Remarkably, only 43 genes showed significant changes (q = .05) of 14,360 expressed genes in flight, and 14374 genes in GC (FPKM>1). This represents a remarkably low 0.3% of the entire genome. Thus, the space flight effect in testis is very focused to a small subset of the genome. In the flight group, elevated expression of testicular and epididymis-specific genes included LCN8 (lipocalin 8) and Defb41, involved in testicular exocrine secretion; Cst11 (cystatin11), responded for gonadal development; Cldn2 (claudin 2) relevant to the blood-testes and blood-epididymal barrier; RNase10, required for post-testicular sperm maturation and male fertility. Significantly down-regulated genes included FABP4-fatty acid binding protein 4, androgen-dependent CFD-complement factor D, Car3, carbonic anhydrase 3, inflammatory marker IL-12. Quantitative RT PCR confirmed dramatic down-regulation of FABP4 (over 20 fold), CFD (5- fold), and

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Car3 (2-fold) in the flight group testis. Testis estrogen receptor-alpha expression was also reduced by μG exposure (P<0.05). We confirmed a 4-fold increased Defb41 in flight vs GC. Differentially expressed genes involved in several canonical pathways included immune responses, metabolic, and endocrine functions.
The mice were sexually mature at launch, so that any observed changes can be attributed to space-flight induced changes in mature testis function and not alterations in a) ongoing maturation of post-natal or b) pre-pubertal testis development, or c) continuing sexual maturation. As such, these animals would be a rodent analogue to adult male astronauts. The RNA quality used from the replicate mice testes from all groups was in excellent condition with no evidence of degradation. The effect of space flight was limited to an extremely small subset of the entire expressed genome with, using high stringency, significant changes in gene expression of only 43 genes out of ~14,370 detected genes. This represents a remarkable 0.3% of the entire genome. This means that the space flight effect is very focused to a small subset of genes in the testis. The strength in this finding is that any pathways identified have a high likelihood of being directly related to space flight effects. This is a valid conclusion since the animals in both flight and BION hardware ground control groups were no different, and unremarkable with respect to body weight (nutrition) and health post-flight. This conclusion is supported by the fact that they were also not significantly different by the same parameters from the parallel vivarium controls that were used as an internal reference standards since the flight and ground control BION groups were asynchronous.

Key observations and their significance (expanded preliminary data analysis): There is a remarkably and highly significant “hot” cluster of interacting physiologic pathways involved in immune regulation. With respect to male reproductive health, this finding is significant since the testis is normally an immunologically privileged site. This immune privilege is maintained by the blood-testis barrier (BTB). The BTB is very similar to the blood-brain barrier in both structure and function. In normal testis, the inflammatory/immune response genes are critical for maintenance of spermatogenesis. The finding shows that space flight significantly affects these pathways. Two preliminary conclusions can be inferred from the initial analysis of the data: 1) space flight has elicited an effect on maintenance of spermatogenesis. Another, equally viable conclusion is that space flight has compromised the blood-testis barrier and the changes are due to mounting of an inflammatory response that will compromise spermatogenesis due to a breach of the blood-testis barrier. Long-established ground-based data has demonstrated that if the BTB has been breached then spermatogenesis will eventually decline and cause infertility. If a true immune response has been raised, then sperm autoantibodies will likely be generated, also causing infertility. Both of these conditions are common causes of human male factor infertility. Further analysis of these samples is required to distinguish between these potential conclusions, including analyzing for the presence of macrophage invasion, and other biomarkers of normal versus exacerbated immune reaction.

Sperm motility assessment showed that swimming velocity is significantly increased in sperm that developed in 30 days of space flight. This is an interesting finding since it has been previously shown that bovine sperm swim faster during brief exposure to microgravity in sounding rocket experiments. Why sperm collected from animals that were exposed to space flight conditions for 30 days also show higher swimming velocity will require additional analysis in future space flight experiments using rodents.

The testis, as the key organ of the male reproductive system has the double function of spermatogenesis and steroidogenesis. Space flight induced a nearly 3-fold increase in the percent of abnormal seminiferous tubules in mice, from 10.5 ± 1.7% in ground controls to 26.2 ± 4.6% in flight animals (P<0.003), including spermatogenic layer disorganization, early release of spermatids and changes of testosterone-producing Leydig cell morphology. An decreased presence of CD68 positive cells has been noted in flight testis Leydig cells and some Sertoli cells compared to GC testis, and in the testis interstitium of flight group. Testis is considered an immunoprivileged organ; it can orchestrate immune responses. Histopathology of the testis suggests that the immune/inflammatory response is related to the observed network pathway induced changes by space flight. One obvious question is whether similar flight-induced pathway changes occurred in other tissues including thymus, bone and muscle.

These findings are also of significant impact with respect to the well-known negative impact of space flight on the immune system in general. One obvious question is whether similar flight-induced pathway changes occurred in other tissues such as thymus (important for immune response) and other tissues where immune responses could elicit space-flight effects such as bone and muscle. These immune pathways all have the H2Aa, H2Ab1, and H2Eb1 genes in common. The tightness of the cluster also highlights the significance of the interactions of the pathways in the space flight effect.

The preliminary data analysis identified the following interacting genes that were differentially regulated in the testis of space mice vs. ground mice. These genes are known to interact highly. These genes are part of the HLA gene family, which encode a group of related proteins known as the human leukocyte antigen (HLA) complex. The HLA complex helps the immune system distinguish the body's own proteins from proteins made by foreign invaders such as viruses and bacteria.

It must be stressed that this initial pathway analysis of the data is limited to identification of the pathways affected and how they relate to each other. We still need to 1) independently verify the changes by RT-PCR, and 2) determine the physiologic consequence of the changes with respect to stimulated or inhibited output of the pathway. This requires a more detailed analysis of the pathways and the component genes to determine the net change in regulatory pathway signaling and biomarker outcomes. For example, if stimulatory genes are upregulated and inhibitory genes are down regulated within a pathway, then the net change will be to stimulate that pathway. Conversely, if the stimulatory genes are down regulated, and the inhibitory genes are upregulated, then the net change in the pathway will be inhibited. The physiologic consequences of such interactions are particularly important for gene products that directly interact as functional regulators.

From the preliminary Upstream Analysis in Ingenuity Pathway Analysis the following regulators and signal transduction molecules and hormones are predicted to be involved with our highly interacting gene list. Additional analysis is needed to ascertain if the signaling pathways involving these regulatory pathways were upregulated or down-regulated by space flight. The results of such analysis will elucidate the physiologic consequences of the changes with respect to testicular function. The Ingenuity Pathway Analysis is using a more relaxed cutoff as some of these genes found in these pathways don’t meet the 1.2 fold q<.05 requirements. However, they are still significant by p-value and really just miss the cutoff. When using a relaxed cutoff list of genes in NCBI’s David (an alternative method for pathway analysis) the same pathways come up as significant for the kegg pathways, and this is again based on the predominant identified HLA genes affected by space flight.

	The testis, as the key organ of the male reproductive system has the double function of spermatogenesis and steroidogenesis. Space flight significantly increased the % of abnormal seminiferous tubules in mice (P<0.01), including spermatogenic layer disorganization, early release of spermatids and abnormal morphology of testosterone-producing Leydig cells. CD68 positive cells were found in both GC and flight group testes. However, in flight animals, CD68-containing macrophages were present exclusively in the interstitial compartment. NextGen comparison of gene expression profiles between flight and GC showed only 0.3% of the entire genome was significantly changed, indicating that the space flight effect is focused on a small subset of genes in the testis. Up-regulated genes are mostly reproductive tissue-specific, but with known impacts in other systems. Down-regulated genes commonly involved in metabolic and inflammatory control, express in different organs, and also down-regulated in STS-131 flight group mouse uteri. Ingenuity Pathway Analysis revealed that significant interacting pathways induced in testis by space flight are involved in immune/inflammatory system regulation and response. Testis is an immunoprivileged organ--it can orchestrate immune responses by regulating the balance of pro- and anti-inflammatory cytokine expression in the Sertoli, Leydig, and peritubular cells. The histopathology of the testis in flight is consistent with alterations in the immune/inflammatory response pathways changes induced by space flight. The increase in inflammatory response and alterations in the Leydig cell population suggest that long term space flight will compromise testis function. In addition to the major foci of immune regulatory pathways, other gene changes were found that have essential regulatory functions on systems whose phenotype would show space-flight effects in other systems include wound healing, cancer, bone and muscle, and endocrine regulation. It should be noted that our findings in testis of male mice from BION were matched with identical changes of the same genes in uteri of female mice from STS-131 and STS-135. Key genes that were significantly decreased in both the male and female in space flight included: FAB4 (inflammatory response, multiple organ neoplasias), CFD (metabolic syndrome, insulin resistance, renal and urologic disorders), CA3 (endometrial cancers, skeletal muscle metabolism), and Cd74 (immune response). Of particular significance is the reduction in 3ß hydroxysteroid dehydrogenase (3ß-HSD). We found over a 50% reduction in the expression of this gene (P < 0.05) in space flight compared to ground controls. 3ß-HSD is a critical gatekeeper enzyme in the synthesis of all the major metabolic and reproductive steroid hormones including progesterone, glucocorticoids, mineralocorticoids, androgens, and estrogens. The decline in the expression of this gene product is consistent with the much wider physiologic decline in multiple organ systems observed in space flight that are regulated by these steroid hormones including bone, muscle, cardiovascular, immune system, metabolism, regulation of intracranial pressure, as well as the reproductive system.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
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